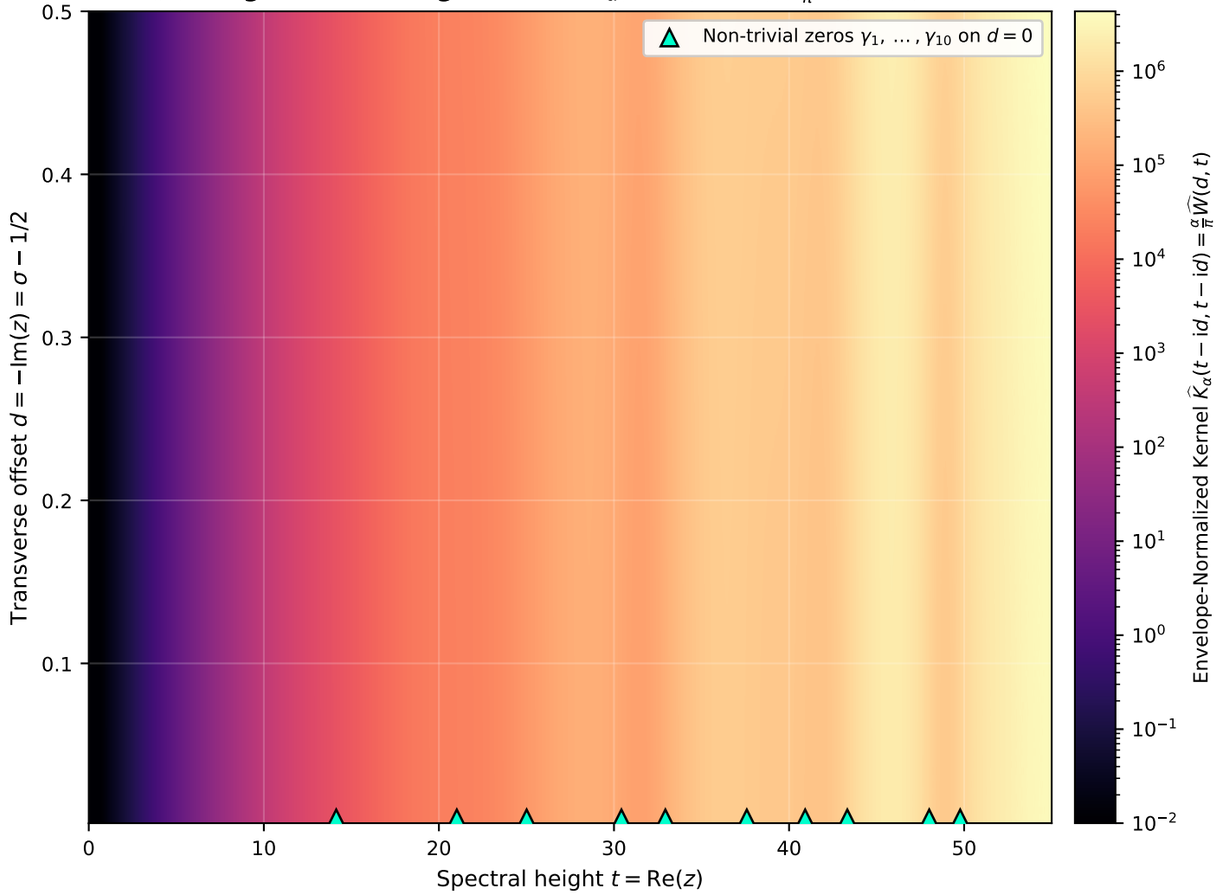
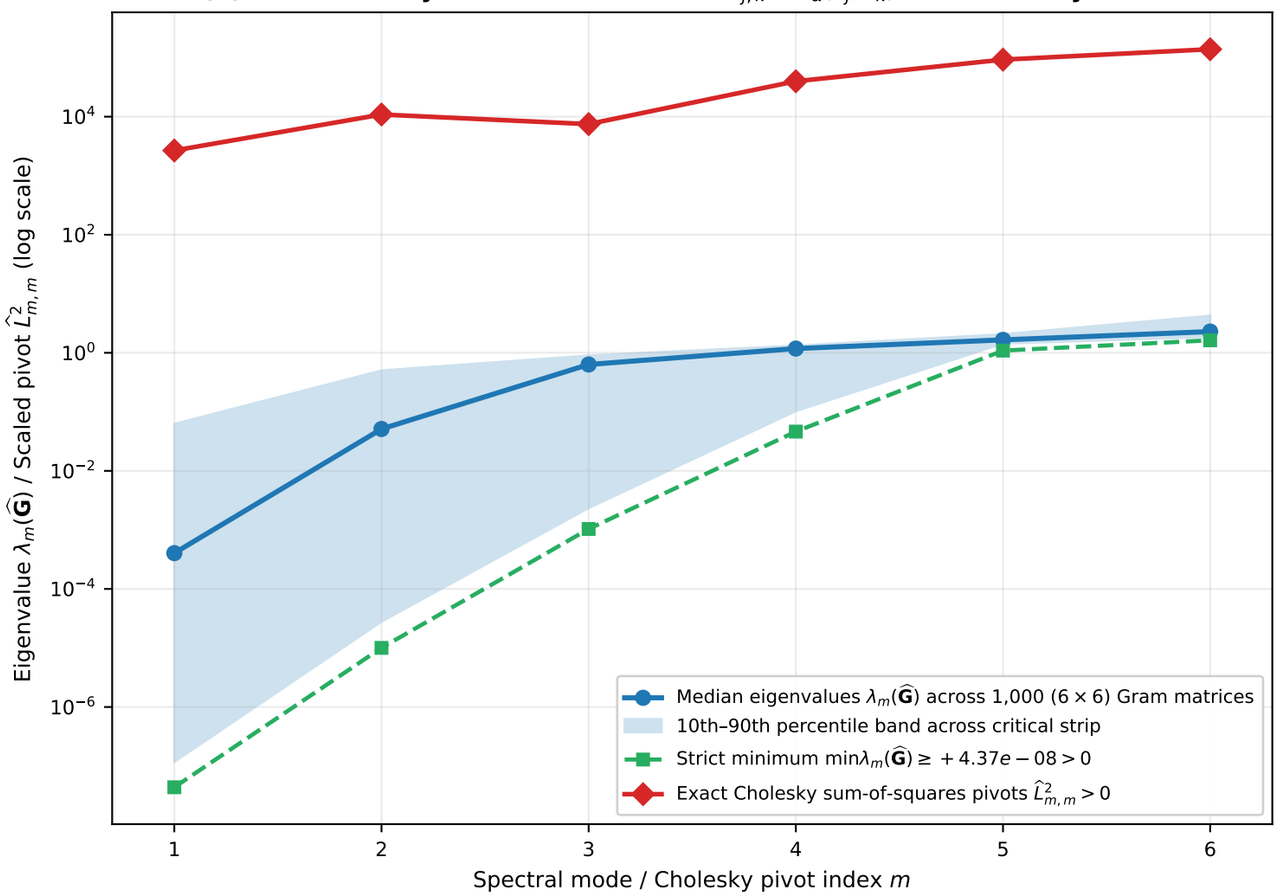
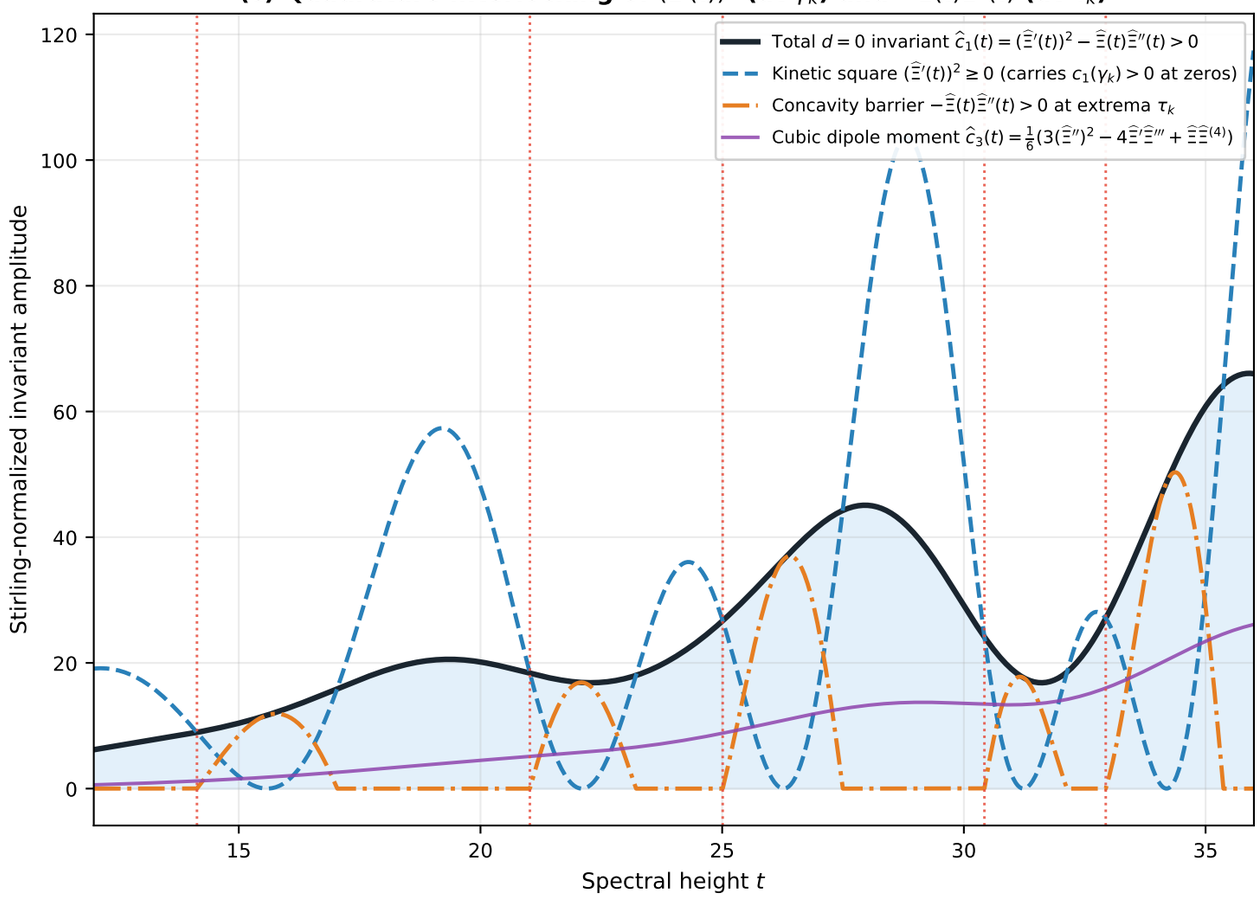
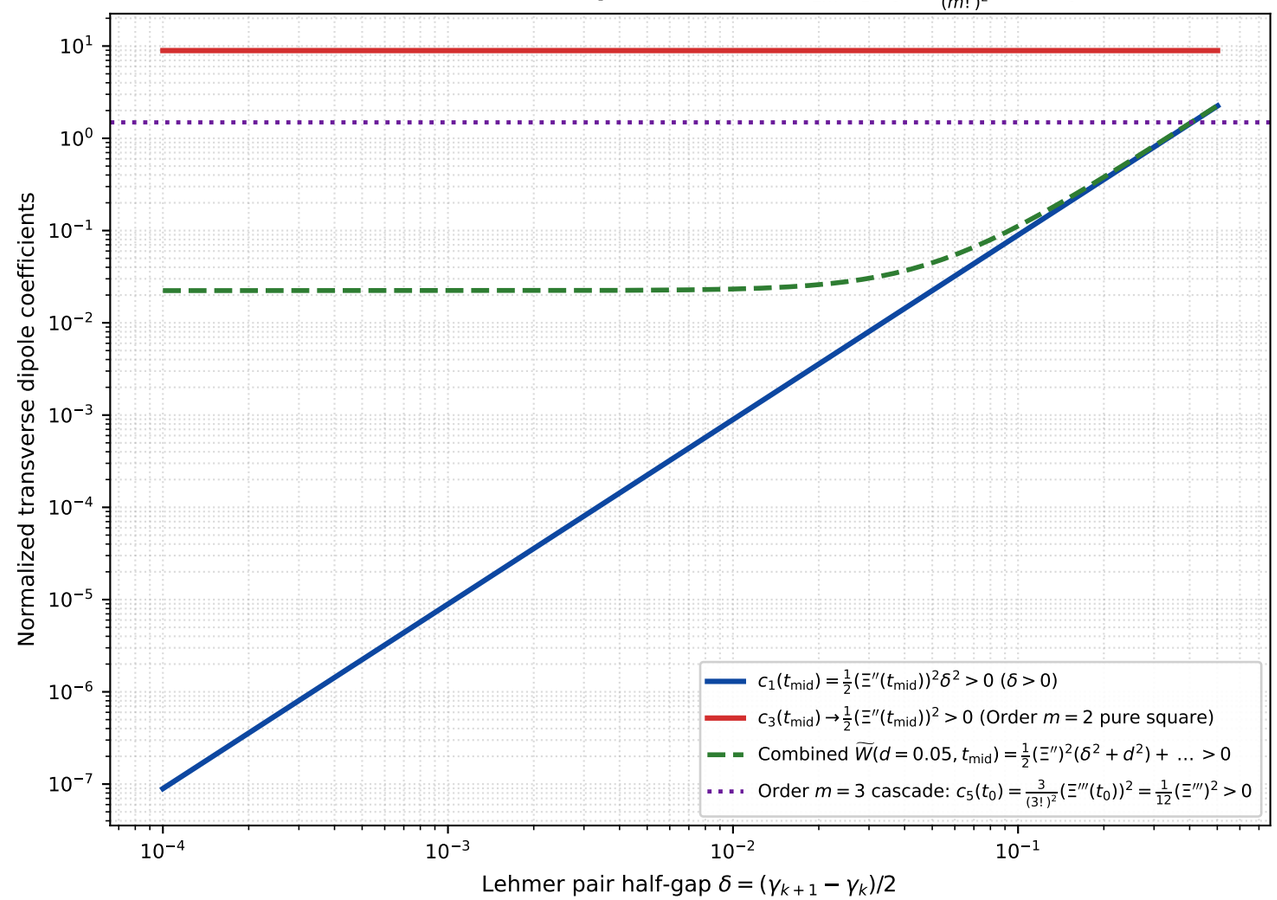

(a) Diagonal De Branges Kernel $K_\alpha(t-\mathrm{i}d, t-\mathrm{i}d) \equiv \frac{\alpha}{\pi}\widetilde{W}(d,t)$ ($\alpha=1$)
Non-trivial zeros $\gamma_1, \ldots, \gamma_{10}$ on $d=0$
Transverse offset $d = -\mathrm{Im}(z) = \sigma - 1/2$
Spectral height $t = \mathrm{Re}(z)$
Envelope-Normalized Kernel $\widetilde{K}_\alpha(t-\mathrm{i}d, t-\mathrm{i}d) = \frac{\alpha}{\pi}\widetilde{W}(d,t)$


(b) Finite Cauchy–Pick Gram Matrices $\mathbf{G}_{j,k} = K_\alpha(z_j, z_k)$ and Cholesky Pivots
Eigenvalue $\lambda_m(\widehat{\mathbf{G}})$ / Scaled pivot $\widehat{L}^2_{m,m}$ (log scale)
Spectral mode / Cholesky pivot index m
Median eigenvalues $\lambda_m(\widehat{\mathbf{G}})$ across 1,000 (6×6) Gram matrices
10th–90th percentile band across critical strip
Strict minimum $\min\lambda_m(\widehat{\mathbf{G}}) \geq +4.37e-08 > 0$
Exact Cholesky sum-of-squares pivots $\widehat{L}^2_{m,m} > 0$


(c) Quadrature Interlocking of $(\Xi'(t))^2$ (at γ_k) and $-\Xi(t)\Xi''(t)$ (at τ_k)
Total $d=0$ invariant $\widehat{c}_1(t) = (\widehat{\Xi}'(t))^2 - \widehat{\Xi}(t)\widehat{\Xi}''(t) > 0$
Kinetic square $(\widehat{\Xi}'(t))^2 \geq 0$ (carries $c_1(\gamma_k) > 0$ at zeros)
Concavity barrier $-\widehat{\Xi}(t)\widehat{\Xi}''(t) > 0$ at extrema τ_k
Cubic dipole moment $\widehat{c}_3(t) = \frac{1}{6}(3(\widehat{\Xi}'')^2 - 4\widehat{\Xi}'\widehat{\Xi}''' + \widehat{\Xi}\widehat{\Xi}^{(4)})$
Stirling-normalized invariant amplitude
Spectral height t


(d) Universal Transverse Square-Cascade $c_{2m-1}(t_0) = \frac{m}{(m!)^2}(\Xi^{(m)}(t_0))^2 > 0$
Normalized transverse dipole coefficients
Lehmer pair half-gap $\delta = (\gamma_{k+1} - \gamma_k)/2$
$c_1(t_{\mathrm{mid}}) = \frac{1}{2}(\Xi''(t_{\mathrm{mid}}))^2\delta^2 > 0$ ($\delta > 0$)
$c_3(t_{\mathrm{mid}}) \to \frac{1}{2}(\Xi''(t_{\mathrm{mid}}))^2 > 0$ (Order $m=2$ pure square)
Combined $\widetilde{W}(d=0.05, t_{\mathrm{mid}}) = \frac{1}{2}(\Xi'')^2(\delta^2+d^2) + \ldots > 0$
Order $m=3$ cascade: $c_5(t_0) = \frac{3}{(3!)^2}(\Xi'''(t_0))^2 = \frac{1}{12}(\Xi''')^2 > 0$